

Fix printer issue

In your Canon iR 2002 printer, the correct terms are **fuser** and **thermistor**.

- **Fuser (fixing unit):** The hot assembly that melts and presses toner powder permanently onto the paper. It normally contains a heater, fixing film or hot roller, pressure roller, thermistor, and safety thermostat.
- **Thermistor:** A small temperature sensor touching or positioned very close to the fuser's heated surface. It continuously reports the fuser temperature to the printer controller.

The control process is:

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1 | Heater warms fuser → Thermistor measures temperature  
2 | → Controller regulates heater → Toner bonds to paper
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For your **E0002/E002 error**, the controller believes the fuser is not heating correctly or cannot maintain the expected temperature. A dirty or defective thermistor can report an incorrect temperature even when the heater itself is working.

The fuser can be extremely hot and carries dangerous electrical voltage. Allow the printer to cool and have a copier technician inspect it rather than touching or bypassing the thermistor.

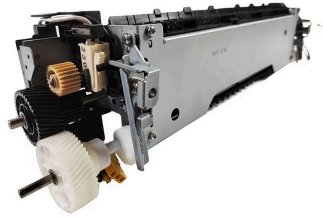
can you give me a picture of the fuser, thermistor of cannon 2002 printer?

Worked for 26s

Yes. These images show the parts used in the **Canon imageRUNNER 2002**.

Complete fuser/fixing unit





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The fuser is the large removable assembly. For many 220–240 V iR 2002 machines, common assembly numbers include **FM1-F160-000** and **FM1-F163-000**, but verify the number printed on your existing unit before purchasing.

Thermistor inside the fuser



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The **thermistor is a small temperature sensor inside the fuser**, touching or sitting extremely close to the fixing film/heater surface. It is difficult to see until the fuser housing is disassembled.

For your **E002 error**, a technician should examine:

- Thermistor surface for baked-on toner
- Thermistor wiring and connector
- Ceramic heater continuity
- Thermal fuse or thermostat
- Fixing film condition

Allow the machine to cool completely and unplug it before opening anything—the fuser contains **high temperature and mains voltage**.

then how to cleanup the fuser?

You can clean the Canon iR 2002 fuser, but only the accessible surfaces. Do not disassemble the fuser unless you are trained—the heater uses mains voltage and becomes extremely hot.

Safe cleaning procedure

1. Turn off the printer with the main power switch.
2. Unplug the power cable.
3. Wait at least **30–45 minutes** for the fuser to cool.
4. Open the rear/right cover and locate the fixing unit in the paper-exit path.
5. Remove loose paper scraps carefully.
6. Wipe accessible paper guides and rollers using:
 - A soft, lint-free cloth
 - Slightly dampened with clean water
 - A dry lint-free cloth afterward
7. Rotate only rollers that turn easily. Never force them.
8. Let everything dry completely, close the covers, and reconnect power.

Cleaning the thermistor

Cleaning the thermistor requires removing and opening the fuser. This is normally technician work:

- Remove the fuser assembly.
- Open its housing.
- Locate the small thermistor contacting the fixing film/heater area.
- Carefully remove baked toner using a lint-free swab and a small amount of **90%+ isopropyl alcohol**.
- Do not bend, scratch, reposition, or press hard on the sensor.
- Inspect the thermistor cable and connector before reassembly.

Do not do these

- Do not use water inside the opened fuser.
- Do not scrape the thermistor with a screwdriver or blade.
- Do not clean the fixing film with abrasive material.
- Do not touch the heater element.
- Do not bypass the thermostat or thermal fuse.
- Do not repeatedly clear the E002 code without fixing its cause.

Cleaning helps only when toner contamination causes an incorrect temperature reading. If **E002 returns**, the ceramic heater, thermistor, thermal fuse, wiring, power supply, or complete fuser may need replacement. A damaged fixing film—wrinkled, torn, or heavily coated—usually requires replacement rather than cleaning.